



**Georgia Health
Sciences University**

Soon to be **Georgia Regents University**

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Laser Safety Guide

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Upon publication of this guide the Laser Safety Subcommittee of the Radiation Safety Committee is hereby authorized to act as agent for the Georgia Health Sciences University (GHSU) in matters of review and control of laser systems on campus. A delineation of responsibilities and authority is included in the following pages.

Furthermore, the GHSU Radiation Safety Office derives its authority directly from the Office of the President of GHSU in all matters involving laser safety or violations of accepted rules of practice as described herein. The Laser Safety Officer is hereby granted authority to immediately suspend laser work that is found to threaten health, property, the environment, or regulatory compliance.

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President

Date

DEFINITIONS

Absorption: Transformation of energy to a different form of energy by interaction with matter.

Accessible emission limit (AEL): The maximum accessible emission level permitted within a particular laser hazard class.

Accessible optical radiation: Optical radiation to which human eye or skin may be exposed for the condition (operation, maintenance or service) specified.

Alpha max: The angular limit subtense of an extended source beyond which additional subtense does not contribute to the hazard and need not be considered. This value is 100 mrad for retinal thermal effects and 110 mrad for retinal photochemical effects.

Alpha min: The angular subtense of a source below which the source can be effectively considered as a point source. The value of alpha min is 1.5 mrad.

Aperture: An opening, window or lens through which radiation can pass.

Attenuation: The decrease in the radiant flux as it passes through an absorbing or scattering medium.

Average power: The total energy in an exposure or emission divided by the duration of exposure or emission.

Aversion response: Closure of the eyelid, eye movement, pupillary constriction or movement of the head to avoid an exposure to a noxious or bright light stimulant. The aversion response to an exposure from a bright, visible, laser source is assumed to occur within 0.25 seconds, including the blink reflex time.

Beam: A collection of light/photonic rays characterized by direction, diameter (or dimensions) and divergence (or convergence).

Beam diameter: The distance between diametrically opposed points in that cross-section of a beam where the power per unit area is $1/e$ (0.368) times that of the peak power per unit area.

Blink reflex: The involuntary closure of the eyes as a result of stimulation by an external event such as irritation of the cornea or conjunctiva, a bright flash, the rapid approach of an object, an auditory stimulus or with facial movements. The ocular aversion response for a bright flash of light is assumed to limit the exposure of a specific retina to 0.25 s or less.

Carcinogen: An agent potentially capable of causing cancer.

Coherent: A beam of light characterized by a fixed phase relation (spatial coherence) or single wavelength, i.e., monochromatic (temporal coherence).

Collateral radiation: Any electromagnetic radiation, except laser radiation, emitted by a laser or laser system, which is physically necessary for its operation.

Collimated beam: Effectively, a “parallel” beam of light with very low divergence or convergence.

Continuous wave (CW): The output of a laser, which is operated in a continuous rather than a pulsed mode. A laser operating with a continuous output for a period > 0.25 sec. is regarded as a CW laser.

Controlled area: An area where the occupancy and activity of those within is subject to control and supervision for the purpose of protection from laser radiation hazards.

Cornea: The transparent outer layer of the human eye, which covers the iris and the crystalline lens. The cornea is the main refracting element of the eye.

Diffraction: Deviation of part of a beam, determined by the wave nature of radiation and occurring when the radiation passes the edge of an opaque obstacle. ?

Diffuse reflection: Change of the spatial distribution of a beam of radiation when it is reflected in many directions by a surface or by a medium.

Divergence: The increase in the diameter of the laser beam with distance from the exit aperture, based on the full angle at the point where the irradiance (or radiant exposure for pulsed lasers) is $1/e$ times the maximum value. Symbol: Φ

Electromagnetic radiation: The flow of energy consisting of orthogonally vibrating electric and magnetic fields lying transverse to the direction of propagation. X-ray, ultraviolet, visible, infrared, and radio waves occupy various portions of the electromagnetic spectrum and differ only in frequency, wavelength, or photon energy.

Embedded laser: An enclosed laser that has a higher classification than the laser system in which it is incorporated, where the system's lower classification is appropriate due to the engineering features limiting accessible emission.

Enclosed laser: A laser that is contained within a protective housing of itself or of the laser or laser system in which it is incorporated. Opening or removal of the protective housing provides additional access to laser radiation above the applicable MPE than possible with the protective housing in place. (An embedded laser is an example of one type of enclosed laser.)

Energy: The capacity for doing work. Energy content is commonly used to characterize the output from pulsed lasers, and is generally expressed in joules (J).

Eye-safe laser: A Class 1 laser product. The use of term Eye-safe laser is discouraged.

Glaucoma: An ocular disease characterized by loss of visual field from damage to the optic nerve and retinal nerve fibers, generally associated with increased intraocular pressure.

Hertz (Hz): A unit expressing the frequency of a periodic oscillation in cycles per second.

Federal Laser Product Performance Standard (FLPPS): The Center for Devices and Radiological Health (CDRH) is a regulatory bureau within the U.S. Federal Food and Drug Administration (FDA) of the Department of Health and Human Services. CDRH has been chartered by Congress to standardize the performance safety of manufactured laser products. All laser products that have been manufactured and entered into commerce, after August 2, 1976, must comply with these regulations. The regulation is known as the Federal Laser Product Performance Standard (FLPPS), and is identified as 21CFR subchapter parts 1040.10 and 1040.11.

There are three Product Performance Schemes:

1. FLPPS/CDRH 21 CFR 1010 and 1040: For countries that recognize U.S. FLPPS. This should be considered if the laser product is only sold in the U.S.

2. IEC 60825 – 1: For countries that only recognize IEC. For example Europe normally only accepts this format.

3. FLPPS/CDRH + Laser Notice # 50: Recommended scheme for selling to an international market. For example selling both into Europe and the U.S. this option should be used. Laser Notice #50 provides guidance on the conditions under which laser product manufacturers may introduce into United States that comply with the IEC standards 60825-1 and IEC 60601-2-22.

Infrared radiation: Electromagnetic radiation with wavelengths that lie within the range 0.7 mm to 1 mm.

Ionizing radiation: Electromagnetic radiation having a sufficiently large photon energy to directly ionize atomic or molecular systems with a single quantum event.

Irradiance (at a point of a surface): Radiant power incident per unit area upon a surface. Unit: W.cm⁻².

Joule (J): A unit of energy. 1 Joule = 1 Watt. second.

Laser: A device that produces radiant energy predominantly by stimulated emission. Laser radiation may be highly coherent temporally, or spatially, or both. LASER is an acronym for Light Amplification by Stimulated Emission of Radiation.

Laser diode: A laser employing a forward-biased semiconductor junction as the active medium.

Laser pointer: A laser product that is usually hand held that emits a low-divergence visible beam and is intended for designating specific objects or images during discussions, lectures or presentations as well as for the aiming of firearms or other visual targeting practice. These products are normally Class 1, Class 2 or Class 3R.

Laser safety officer (LSO): One who has authority and responsibility to monitor and enforce the control of laser hazards and effect the knowledgeable evaluation and control of laser hazards.

Laser system: An assembly of electrical, mechanical, and optical components, which includes a laser.

Lesion: An abnormal change in the structure of an organ or part due to injury or disease.

Limiting aperture diameter: The diameter of a circle over which irradiance or radiant exposure is averaged for purposes of hazard evaluation and classification. Symbol: Df

Magnified Viewing: Viewing a small object through an optical system that increases the apparent object size. This type of system can make a diverging laser beam more hazardous (e.g., using a magnifying optic to view an optical fiber with a laser beam emitted).

Maximum permissible exposure (MPE): The level of laser radiation to which an unprotected person may be exposed without adverse biological changes in the eye or skin.

Meter: A unit of length in the international system of units; currently defined as the length of a path traversed in vacuum by light during a period of $1/299792488$ seconds. Typically, the meter is subdivided into the following units:

centimeter (cm) = 10^{-2} m
millimeter (mm) = 10^{-3} m
micrometer (mm) = 10^{-6} m
nanometer (nm) = 10^{-9} m

Minimum viewing distance: The minimum distance at which the eye can produce a focused image of a diffuse source, usually assumed to be 10 cm.

Nominal hazard zone (NHZ): The space within which the level of the direct, reflected or scattered radiation may exceeds the applicable MPE. Exposure levels beyond the boundary of the NHZ are below the applicable MPE .

Non-beam hazard: A class of hazards that result from factors other than direct human exposure to a laser beam.

Optically aided viewing: Viewing with a telescopic (binocular) or magnifying optic. Under certain circumstances, viewing with an optical aid can increase the hazard from a laser beam (see telescopic viewing or magnified viewing).

Optical density: Logarithm to the base ten of the reciprocal of the transmittance: $DI = \log_{10}(1/t_l)$, where t_l is transmittance at the wavelength of interest.

Photochemical effect: A biological effect produced by a chemical action brought about by the absorption of photons by molecules that alter the molecule.

Plasma radiation: Black body radiation generated by luminescence of matter in a laser-generated plume.

PLU: Principal Laser User

Power: The rate at which energy is emitted, transferred, or received. Unit: watts (joules per second).

Protective housing: An enclosure that surrounds the laser or laser system that prevents access to laser radiation above the applicable MPE level. The aperture through which the useful beam is emitted is not part of the protective housing. The protective housing may enclose associated optics and a work station and shall limit access to other associated radiant energy emissions and to electrical hazards associated with components and terminals, and may enclose associated optics and a workstation.

Pulse duration: The duration of a laser pulse; usually measured as the time interval between the half-power points on the leading and trailing ends of the pulse.

Pulsed laser: A laser, which delivers its energy in the form of a single pulse or a train of pulses. In the ANSI standard, the duration of a pulse is < 0.25 sec.

Pupil: The variable aperture in the iris through which light travels to the interior of the eye.

Q-switch: A device for producing very short ($\gg 10 - 250$ ns), intense laser pulses by enhancing the storage and dumping of electronic energy in and out of the lasing medium, respectively.

Q-switched laser: A laser that emits short ($\gg 10 - 250$ ns), high-power pulses by means of a Q-switch.

Radian (rad): A unit of angular measure equal to the angle subtended at the center of a circle by an arc whose length is equal to the radius of the circle. 1 radian $\gg 57.3$ degrees, 2π radians = 360 degrees.

Radiance: Radiant flux or power output per unit solid angle per unit area. Unit: Watts per centimeter squared per steradian ($W \cdot cm^{-2} \cdot sr^{-1}$). Symbol: L

Radiant energy: Energy emitted, transferred, or received in the form of radiation. Unit: joule (J).

Radiant exposure: Surface density of the radiant energy received. Unit: joules per centimeter squared ($J \cdot cm^{-2}$).

Radiant flux: Power emitted, transferred, or received in the form of radiation. Unit: watt (W). Also called radiant power. Symbol: Φ

Reflectance: The ratio of the total reflected radiant power to the total incident power, also called “reflectivity.”

Reflection: Deviation of radiation following incidence on a surface.

Repetitive pulsed laser: A laser with multiple pulses of radiant energy occurring in sequence with a pulse repetition frequency (prf) ≥ 1 Hz.

Retina: The sensory membrane, which receives the incident image formed by the cornea and lens of the human eye. The retina lines the inside of the eye.

Safety latch: A mechanical device designed to require a conscious decision to override the latch to gain entry to a controlled area.

Specular reflection: A mirror-like reflection.

Steradian (sr): The unit of measure for a solid angle. There are 4π steradians about any point in space.

Telescopic viewing: Viewing an object from a long distance with the aid of an optical system that increases the size of the image. The system (e.g., binoculars) generally collects light through larger aperture thus magnifying hazards from large-beam, collimated lasers.

Transmission: Passage of radiation through a medium.

Transmittance: The ratio of transmitted power (energy) to incident power (energy).

Ultraviolet radiation: Electromagnetic radiation with wavelengths between 0.18 to 0.4 μm (shorter than those of visible radiation).

Visible radiation (light): Electromagnetic radiation, which can be detected by the human eye. This term is commonly used to describe wavelengths that lie in the range 0.4 to 0.7 μm .

Watt: The unit of power or radiant flux. 1 Watt = 1 Joule-per-second.

Wavelength: The distance in the line of advance of a sinusoidal wave from one point to the next point of corresponding phase (e.g., distance from one peak to the next).

1.0 Introduction.

The primary objective of the GHSU Laser Safety Program is to ensure the safe use of laser radiation and to assure no laser radiation in excess of the maximum permissible exposure (MPE) limit reaches human eyes or skin. Additionally, the program is designed to ensure that adequate protection against collateral hazards is provided (collateral hazards are described in Section 5.0 of this guide). See Laser Safety Program Policy (Appendix A)

This guide summarizes the requirements and responsibilities of laser users at Georgia Health Sciences University (GHSU). PLUs, (Principal Laser User) and staff affected by laser operations can have access to the Laser Safety Guide online at Radiation Safety's website at (<http://georgiahealth.edu/services/ehs/radsafe>). Laser users are required to familiarize themselves with the provisions of this guide.

GHSU uses laser systems in education, research, and patient care. The Laser Safety Subcommittee (LSS) may authorize staff or faculty members as Principal Laser Users after a review of proposed use, adequacy of facilities, and training and experience of the applicant.

To achieve this objective, GHSU requires that all Class 3B and Class 4 lasers and laser systems operate in accordance with the American National Standards Institute (ANSI) Z136.1-2007 Safe Use of Lasers, (ANSI) Z136.5-2009 Safe Use of Lasers in Educational Institutions, (ANSI) Z136.3-2011 Safe Use of Lasers in Health Care and applicable federal and state regulations. The Laser Safety Program applies to all GHSU locations, including off-campus, mobile and temporary locations.

2.0 Responsibilities.

2.1 Laser Safety Subcommittee (LSS). The LSS of the Radiation Safety Committee (RSC) is an advisory body for all aspects of laser safety and use at GHSU. The LSS may grant, deny, or withdraw authorization for laser use. The LSS is responsible for:

- 2.1.1 Ensuring GHSU procedures, safety guides, criteria for training, and testing of each category of workers involved with the use of laser devices are implemented according to accepted safety practices.
- 2.1.2 Assuring that only qualified individuals are permitted to use laser devices or supervise such use by others.
- 2.1.3 Reviewing annual reports from the Laser Safety Officer (LSO) summarizing laser inspections, including compliance with safety related engineering and administrative controls and use of personal protective equipment, occupational exposure for all personnel, and compliance with Federal/State of Georgia regulations.

- 2.1.4 Enforcing compliance with the program, including imposition of sanctions for noncompliance.
 - 2.1.5 Providing advice to research groups, departments and investigators via the LSO.
 - 2.1.6 Advising on the need for medical surveillance.
 - 2.1.7 Tracking corrective actions for items of concern identified by the LSO or the LSS.
- 2.2 Laser Safety Officer (LSO). The LSO is responsible for the day to day administration and implementation of the procedures that support safety and compliance in the Laser Safety Program. The LSO shall be qualified by completion of recognized industry training and shall be approved by the LSS. The LSO is responsible for:
- 2.2.1 Review and approval of all requisitions for laser systems prior to purchase.
 - 2.2.2 Classifying lasers and laser systems.
 - 2.2.3 Conducting hazard evaluations of laser work areas, including the establishment of Nominal Hazard Zones (NHZ).
 - 2.2.4 Assuring that the prescribed control measures are implemented and maintained. This includes recommending or approving substitute or alternate control measures when the primary ones are not feasible or practical.
 - 2.2.5 Reviewing and recommending approval of Class 3B and Class 4 standard operating procedures.
 - 2.2.6 Recommending and approving protective equipment including eyewear, clothing, barriers, screens, etc., as may be required to assure personnel safety.
 - 2.2.7 Approving the wording on area signs and equipment labels.
 - 2.2.8 Review and approval for new or modified Class 3B and Class 4 laser facilities prior to use.
 - 2.2.9 Assuring that laser use areas and laser equipment are audited periodically to assure proper operation.

- 2.2.10 The LSO shall assure that adequate safety education and training, including refresher training, is provided to laser personnel involved in the use of lasers.
 - 2.2.11 Assuring that the necessary records (required by applicable government regulations/standards maintenance of safety programs, SOPs, audits, etc.) are maintained.
 - 2.2.12 Maintaining an Emergency Response Plan for actual or suspected exposure to potentially harmful laser radiation.
 - 2.2.13 Submit periodic reports to the LSS detailing the status of the Laser Safety Program.
- 2.3 Principal Laser User (PLU). The PLU shall be knowledgeable of the requirements for laser safety; the potential hazards and associated control measures for laser systems; and policies, practices and procedures pertaining to lasers under their authority. The PLU is responsible for:
- 2.3.1 Instructions and training on laser hazards and safety controls for Laser Users under his/her authority.
 - 2.3.2 Supervising Laser Users under his/her authority to ensure safe operation of laser systems.
 - 2.3.3 Ensuring adequate control of hazards to employees, visitors, and the general public prior to allowing the laser to be used.
 - 2.3.4 Ensuring that individuals scheduled to work with lasers are listed on Laser User Registration Forms.
 - 2.3.5 Purchasing laser safety personal protective equipment and devices.
 - 2.3.6 Ensuring that sufficient protective eyewear of the correct optical density is readily available, in good condition, and worn during laser setup and operation.
 - 2.3.7 Implementing the laser accident response plan for known or suspected laser exposures.
 - 2.3.8 Reporting all laser accidents or injuries to the LSO immediately.
 - 2.3.9 Submitting plans for new or modified Class 3B and Class 4 laser installations to the LSO for review and approval.

- 2.3.10 Obtaining LSO approval before placing Class 3B or Class 4 lasers into operation.
 - 2.3.11 Notifying the LSO when decommissioning, transferring or disposing of lasers or laser systems.
 - 2.3.12 Obtaining a baseline optical exam for laser workers before they begin work with Class 3B and Class 4 lasers if required by the LSO.
 - 2.3.13 Developing written laser specific SOPs for Class 3B and Class 4 lasers and laser systems. The SOPs shall include instructions for operation, maintenance, and other relevant safety considerations. (See the SOP template at Appendix B).
 - 2.3.14 Enforcing the safety requirements prescribed in this guide
 - 2.3.15 Laser accident responsibilities.
 - 2.3.15.1 Assist the affected worker in obtaining medical support.
 - 2.3.15.2 Stop work using the laser system and preserve the incident scene for investigation.
 - 2.3.15.3 Notify the Laser Safety Officer (or the on-call Radiation Safety staff member) as soon as possible.
 - 2.3.16 Retaining laser safety-related records, including laser safety training for each employee for a minimum of one year beyond termination of duties. Standing Operating Procedures, inspection reports, operating manuals, and vendor specifications shall be retained for a minimum of one year beyond termination of use. Medical records of all personnel included in the medical surveillance program should be retained for at least 30 years.
- 2.4 Laser Users. Laser Users operate laser systems under the supervision of the PLU. Laser Users are responsible for:
- 2.4.1 Reading and complying with this guide.
 - 2.4.2 Completing laser safety training prior to starting work with lasers.
 - 2.4.3 Obtaining a baseline optical exam before starting work with lasers, if required by the LSO.
 - 2.4.4 Not energizing or working with or within a laser's nominal hazard zone unless authorized to do so by the PLU.

- 2.4.5 Complying with safety rules, procedures, and instructions prescribed by the supervisor and the LSO, to include wearing the appropriate protective eyewear.
- 2.4.6 Understanding laser operating procedures.
- 2.4.7 Restricting access of ancillary services personnel into a room or area where a Class 3B or Class 4 laser is operating.
- 2.4.8 Reporting all laser accidents or injuries to the PLU and the LSO.
- 2.4.9 Obtaining PLU approval before deviating from the SOP.

3.0 Laser System Classification. Lasers produce intense, coherent beams of electromagnetic radiation. Interaction with the human body is generally at the surface, affecting the eye and the skin. Laser classification is related to the potential for biological injury resulting from exposure to the eye (by direct viewing or viewing through a lens), or exposure to the skin.

- 3.1 Class 1 Laser Systems. Any laser or laser system containing a laser that cannot emit accessible radiation levels during operation in excess of the applicable Class 1 Accessible Emission Level (AEL) for any duration within the maximum duration (30,000 s, except for infrared system where 100 s shall be used) inherent in the design or intended use of the laser or laser system. Class 1 laser systems are incapable of producing damaging radiation levels during operation and are exempt from controls or surveillance.
- 3.2 Class 1M Laser Systems. Any laser or laser system containing a laser that cannot emit during operation accessible radiation levels in excess of the applicable Class 1 AEL under the conditions of measurement for the unaided eye, but exceeds the Class 1 AEL for telescopic viewing and does not exceed the Class 3B AEL for any emission duration within the maximum duration (30,000 s) inherent in the design or intended use of the laser or laser system. Class 1M laser systems are incapable of producing hazardous exposure conditions during operation unless the beam is viewed with an optical instrument such as an eye-loupe (diverging beam) or a telescope (collimated beam). Class 1M lasers are exempt from controls other than to prevent potentially hazardous optically aided viewing, and are exempt from other forms of surveillance.
- 3.3 Class 2 Laser Systems are visible continuous wave (CW) and repetitive-pulse laser systems which can emit accessible radiation energy exceeding the appropriate Class 1 AEL for the maximum duration inherent in the design or intended use of the system, but not exceeding the Class 1 AEL for any applicable pulse emission duration < 0.25 s and not exceeding average radiation power of 1 mW. Class 2 lasers emit in the visible portion of the spectrum (0.4 to 0.7 μ m), and eye protection is normally afforded by the aversion response.

3.4 Class 2M Laser Systems. Any laser or laser system that cannot emit during operation accessible radiation levels in excess of the applicable Class 2 AEL under the conditions of measurement for the unaided eye, but exceeds the Class 2 AEL for telescopic viewing and does not exceed the Class 3B AEL, for any emission duration within the maximum duration (0.25 s) inherent in the design or intended use of the laser or laser system. Eye protection is generally afforded by the aversion response, but Class 2M lasers are potentially hazardous when viewed with optical aids.

3.5 Class 3R Laser Systems have an accessible output between 1 and 5 times the Class 1 AEL for wavelengths shorter than 0.4 μm or longer than 0.7 μm , or less than 5 times the Class 2 AEL for wavelengths between 0.4 and 0.7 μm . Class 3R lasers are potentially hazardous under some direct and specular reflection viewing condition if the eye is appropriately focused and stable, but the probability of an actual injury is small. This laser will not pose either a fire hazard or diffuse reflection hazard.

3.6 Class 3B Laser Systems. The AEL for class 3B lasers is greater than that for class 3R lasers and average radiant power is less than or equal to 0.5 W for exposure time greater than or equal to 0.25 s. Radiant energy for class 3B lasers is less than or equal to 0.125 J for UV (0.18 to 0.4 μm) and infrared (1.4 μm to 1 mm) lasers for exposure times less than 0.25 s. Radiant energy limit for class 3B visible (0.4- 0.7 μm) and near infrared lasers (0.7- 1.4 μm) is less than or equal to 0.03 J C_A per pulse. Where $C_A = 10^2 (\lambda - 0.700)$ for wavelength (λ) from 0.700 to 1.050 μm ; and $C_A = 5$ for λ from 1.050 to 1.400 μm .

Class 3B lasers may be hazardous under direct and specular reflection viewing conditions, but are normally not a diffuse-reflection or fire hazard. Class 3B lasers require labeling and physical controls to prevent viewing of direct and reflected beam.

3.7 Class 4 Laser Systems. The AEL for Class 4 lasers and laser systems is greater than that for Class 3B lasers. Class 4 lasers present a hazard to eye and skin from the direct beam and may pose a diffuse reflection hazard. Class 4 lasers may also produce laser generated air contaminants (LGAC) and hazardous plasma radiation. Class 4 lasers require labeling and physical controls to prevent viewing of the direct, reflected, and diffuse reflected beam.

3.8 Embedded Laser Systems. Class 2, 3, or 4 lasers contained in a protective housing and operated in a lower classification mode may be reclassified downward. Specific control measures are required to maintain the lower classification. Approval of the LSO is required for reclassification of embedded systems that are non-commercial design and construction.

4.0 Laser System Acquisition, Transfer, and Disposal are done in accordance with Georgia Department of Public Health Chapter 270-6-27, *Rules and Regulations for Laser Radiation*.

4.1 Laser System Acquisition.

- 4.1.1 Staff or faculty anticipating purchase of a Class 3B or Class 4 laser system shall coordinate with the LSO in advance of purchase.
 - 4.1.2 The LSO shall approve all purchases of Class 3B and Class 4 laser systems.
 - 4.1.3 Upon receipt and prior to use the PLU or prospective PLU) shall complete the laser registration form (Appendix B) and submit it to the LSO.
- 4.2 Laser System Transfer. The PLU shall notify the LSO when a Class 3B or Class 4 laser is to be moved within the facility, or is transferred to another PLU prior to being moved. The laser may not be operated in the new location until the LSO completes the laser hazard evaluation and all required controls are in place.
- 4.3 Laser System Disposal. The PLU shall notify the LSO when a laser system is no longer used, declared as excess property, or designated for disposal.

5.0 Laser System Hazards

5.1 Eye Hazards. The major hazard of laser light is from beams entering the eye. The eye is the organ most sensitive to light. Just as a magnifying glass can be used to focus the sun and burn wood, the lens in the human eye focuses the laser beam into a tiny spot that can burn the retina. A laser beam with low divergence entering the eye can be focused down to a 10 to 20 microns diameter spot at the retina, thereby increasing the irradiance of the beam at the retina by 100,000 to 200,000 times of the beam entering the eye.

5.2 Skin Hazards. Lasers can harm the skin via photochemical or thermal burns. Depending on the wavelength, the beam may penetrate both the epidermis and the dermis. The epidermis is the outermost living layer of skin. Far and mid-ultraviolet (the actinic UV) are absorbed by the epidermis. Sunburn (reddening and blistering) may result from short-term exposure to the beam. UV exposure is also associated with an increased risk of developing skin cancer and premature aging (wrinkles, etc.) of the skin.

Thermal burns to the skin are rare. They usually require exposure to high-energy beams for an extended period of time. Carbon dioxide and other infrared lasers are most commonly associated with thermal burns, since this wavelength range penetrates deeply into skin tissue. The resulting burn may be first degree (reddening), second degree (blistering) or third degree (charring).

Some individuals are photosensitive or may be taking prescription drugs that induce photosensitivity. Particular attention must be given to the effect of antibiotics and fungicides when working around lasers.

- 5.3 Electrical Hazards. Laser systems can present an electric shock hazard. This may occur from contact with exposed utility power, device control, and power supply conductors operating at potentials of 50 volts and above. These exposures can occur during laser setup or installation, maintenance and service, where equipment protective covers are often removed to allow access to active components as required for those activities. Those exposed can be equipment installers, users, technicians, and uninformed members of the public.

The effect upon those who accidentally come into contact with energized conductors at or above 50 volts can range from a minor “tingle”, to startle reaction, to serious personal injury or death. Electric shock is a very serious hazard, and deaths associated with laser systems have occurred.

- 5.4 Fire Hazards. Class 4 laser beams present a fire hazard. Enclosure of Class 4 beams can result in potential fire hazard if the enclosure materials are likely to be exposed to irradiances exceeding 10W per square centimeter or beams exceeding 0.5W. Operators of Class 4 lasers should be aware of the ability of combustible materials such as unprotected wire insulation, plastic, rubber, paper, human tissues, human hair, intestinal gases, and skin treated with acetone and alcohol-based preparation to catch on fire from intense direct or reflected beams. The risk of fire is greater in oxygen-rich atmospheres.

- 5.5 Collateral and Plasma Radiation. Collateral radiation is radiation other than that associated with the primary laser beam and is produced by system components such as power supplies, discharge lamps and plasma tubes. Such radiation may take the form of x-radiation, UV, visible, IR, microwave and radiofrequency (RF) radiation. Plasma radiation is generated when an energetic laser beam interacts with matter, typically metals, when high power pulsed laser beams (peak irradiance of the order of 1012 W/cm² are focused on a target.

- 5.6 Laser Generated Airborne Contaminants. In operations using lasers that vaporize tissue through disruption of cells, laser generated airborne contaminants (LGAC) result as an airborne hazard requiring appropriate management. Analysis of these contaminants produced during laser surgical procedures has shown the presence of gaseous toxic compounds, bio-aerosols, cellular material, and viruses.

At certain concentrations, some of the LGAC may cause ocular and upper respiratory tract irritation, have unpleasant odors, create visual problems for the user, and have been shown to have mutagenic and carcinogenic potential. It has been shown that laser smoke production is a function of increased irradiance levels. Therefore, laser surgical procedures requiring high irradiance levels are more likely to produce LGAC.

6.0 Laser System Safety Measures

6.1 Class 1, 2, and 3R Laser Systems. When used as intended Class 1, 2, and 3R laser systems are generally low hazard devices. The following requirements apply:

- 6.1.1 PLUs are responsible for ensuring training on proper use of that equipment by Laser Users.
- 6.1.2 Exposure to laser radiation must be kept below the Maximum Permissible Exposure (MPE) under all conditions of operation or maintenance.
- 6.1.3 Laser systems must have the appropriate warning labels with the laser sunburst logotype symbol and the appropriate cautionary statement.
- 6.1.4 Removal of protective housing or system modification can increase a laser's classification. See paragraph 11.1.
- 6.1.5 Use of Class 3R laser with telescopes, microscopes, or alignment devices shall be reviewed by the LSO prior to operation.

6.2 Class 3B Laser Systems require a laser hazard analysis, including determination of the MPE and Nominal Hazard Zone (NHZ), by the LSO. The hazard analysis determines the laser control area. Safety measures include:

- 6.2.1 Only authorized personnel (designated PLUs and Laser Users) may operate Class 3B lasers.
- 6.2.2 The controlled area must be posted with the appropriate warning signs. Minors (persons under age 18) are not permitted in the controlled area.
- 6.2.3 All area or entryway safety controls must be designed to allow rapid egress by laser personnel and admittance to the laser controlled area under emergency conditions.
- 6.2.4 The laser beam path shall be well defined and project into a controlled space.
- 6.2.5 Windows, doorway, and open portals shall be covered or restricted in such a manner as to reduce the transmitted laser radiation to levels at or below the applicable ocular MPE.
- 6.2.6 Only diffusely reflecting materials should be in or near the beam path.
- 6.2.7 Appropriate protective eyewear shall be readily available.
- 6.2.8 See paragraph 6.4 for additional requirements and recommendations.

6.3 Class 4 Laser Systems must incorporate all of the safety measures for Class 3B laser systems, in addition to the safety measures listed below:

- 6.3.1 The control area requires interlocks or alternate controls to preclude the entry of unprotected personnel while laser radiation is present. The interlock system may be designed to preclude entry while the laser is operating or to terminate laser operation when the door is opened without deliberate overriding of the interlock, *or*
- 6.3.2 A blocking barrier, screen, or curtains must be used to block, screen, or attenuate the laser radiation levels so that MPE is not exceeded at the entry point, *or*
- 6.3.3 At the entryway there must be a visible or audible signal indicating that the laser is energized and operating at Class 4 levels. A lighted laser warning sign or flashing light (visible through protective eyewear) are acceptable entryway warning light alternatives.

6.4 Engineering Controls for Class 3B and Class 4 Laser Systems. The engineering control measures required for Class 3B and 4 lasers are listed in the table below. Where specific engineering controls are not feasible they may be replaced with specific administrative and procedural controls and personal protective equipment (PPE) with prior approval of the LSO. Alternative controls and PPE requirements must be documented in a written SOP.

Engineering Controls			
☒ Required	☒ Recommended	Class 3B	Class 4
Protective Housing – for active laser work with housing off, contact LSO for hazard analysis and appropriate controls		☒	☒
Interlocks on Protective Housing		☒	☒
Service Access Panels Interlocked or tool required and appropriate warning label on the panel		☒	☒
Key Control		☒	☒
Remote Interlock Connector		☒	☒
Beam stop or Attenuator		☒	☒
Laser Activation Warning System		☒	☒
Emission Delay		NA	☒
Remote Firing and Monitoring		NA	☒
Panic Button		NA	☒
Viewing windows, diffuse display screens, or collecting optics (lenses, microscopes, etc.) are controlled with interlocks, filters, or attenuators to maintain laser radiation at the viewing position at or below the MPE		☒	☒
Enclosed Beam Path		☒	☒

7.0 Laser System Protection.

7.1 Protective Eyewear.

- 7.1.1 Eye protection devices shall be worn when working with Class 3B and Class 4 lasers or laser systems. Laser protective eyewear is usually not required for Class 2 or Class 3A lasers or laser systems, except in conditions where intentional long-term (>0.25 seconds) direct viewing is required. Eyewear must be specifically selected to withstand either direct or diffusely scattered beams and shall meet all provisions of ANSI Z87.1-1989.
- 7.1.2 Eyewear must be inspected before each use, and replaced if necessary, to maintain the eyewear in good condition.
- 7.1.3 Contact the LSO for assistance in selecting protective eyewear. Factors to consider in selecting appropriate eyewear include:
 - 7.1.3.1 Laser power and/or pulse energy.
 - 7.1.3.2 Wavelength(s) of laser output.
 - 7.1.3.3 Potential for multi-wavelength operation.
 - 7.1.3.4 Radiant exposure or irradiance levels for which protection (worst case) is required.
 - 7.1.3.5 Exposure time criteria.
 - 7.1.3.6 Maximum permissible exposure.
 - 7.1.3.7 Optical density requirement of eyewear filters at laser output wavelength.
 - 7.1.3.8 Angular dependence of protection afforded.
 - 7.1.3.9 Visible light transmission requirement and assessment of the effect of the eyewear on the ability to perform tasks while wearing the eyewear.
 - 7.1.3.10 Need for side-shield protection and maximum peripheral vision.
 - 7.1.3.11 Radiant exposure or irradiance and the corresponding time factors at which laser safety filter characteristics change occurs, including transient bleaching especially for ultra-short pulse lengths.

7.1.3.12 Need for prescription glasses.

7.1.3.13 Comfort and fit.

7.1.3.14 Degradation of filter media, such as photo bleaching.

7.1.3.15 Strength of materials (resistance to mechanical trauma and shock).

7.1.3.16 Capability of the front surface to produce a hazardous specular reflection.

7.1.3.17 Requirement for anti-fogging design or coatings.

7.2 Skin Protection.

7.2.1 Skin protection can best be achieved through engineering controls. If potential skin damaging exposures exist, skin covers and/or “sun screen” creams are recommended.

7.2.2 Minimize exposure to UV radiation by using beam shields and clothing (opaque gloves, tightly woven fabrics, laboratory jacket or coat) which attenuate the radiation to levels below the MPE for specific UV wavelengths. Consider flame retardant materials for Class 4 lasers.

7.2.3 Special attention must be given to the possibility of producing undesirable reactions in the presence of UV radiation (formation of skin sensitizing agents, ozone, etc.).

7.3 Facility Windows. Exterior or interior windows that are located within the NHZ of a Class 3B or Class 4 laser must be provided with appropriate absorbing filter, scattering filter, blocking barrier or screen to reduce any transmitted laser radiation to levels below the applicable MPE level. Important factors for selection include: ability to withstand direct and diffusely scattered beams, flammability and decomposition products of the window material.

7.4 Protective Barriers and Curtains. A blocking barrier, screen or curtain which can block or filter the laser beam at the entryway should be used inside the controlled area to prevent Class 3B or Class 4 laser light from exiting the area at levels above the applicable MPE level. Important factors for selection include: ability to withstand direct and diffusely scattered beams, flammability and decomposition products of the protective barrier or curtain.

7.5 LGAC Protection. In general, there are two techniques to protect exposed personnel from LGAC; the use of ventilation and respiratory protection.

- 7.5.1 Ventilation is the most important protective measure and a primary way to control exposures to LGAC. Ventilation can include protection by general room or local exhaust, or a combination. Local exhaust includes smoke evacuators or wall suction systems. Portable smoke evacuators use various filters and absorbers that require monitoring and replacement on a regular basis. These filters should be considered a possible biohazard and should be disposed of accordingly. To obtain maximum capture, the smoke evacuator nozzle should be placed within two inches of the laser site.
- 7.5.2 Respiratory protection is a secondary way to control exposures to LGAC if ventilation is not feasible to control exposures. Adequate respiratory protection is generally afforded by a filtering face piece respirator. Use of respirators requires a respirator medical evaluation, fit test, and training by Occupational Health Services prior to using the respirator. These requirements apply each year thereafter.

8.0 Laser Warning Signs and Labels. Laser controlled areas must be posted with the appropriate warning signs at the entryway(s) and if necessary, within the laser controlled area.

8.1 Warning Signs.

- 8.1.1 Danger: Must be used with all signs and labels associated with all Class 3R lasers and laser systems that exceed the appropriate MPE for irradiance, and all Class 3B and Class 4 lasers and laser systems.
- 8.1.2 Caution: Must be used with all signs and labels associated with Class 2 lasers and laser systems, and all Class 3A lasers and laser systems that do not exceed the appropriate MPE for irradiance.
- 8.1.3 Notice: Must be used on signs posted outside a temporary laser controlled area. The area within the temporary controlled area must also have appropriate signs posted (danger warning for Class 3B or Class 4).

8.2 Laser Equipment Labels. All lasers or laser systems (except Class 1) must have appropriate warning labels affixed to a conspicuous place on both the housing and the control panel (if separated by more than 2 meters).

- 8.2.1 Class 2 lasers and laser systems: "Laser Radiation – Do Not Stare into Beam".
- 8.2.2 Class 3R lasers and laser systems (accessible irradiance does not exceed MPE based upon 0.25 second exposure for wavelengths between 0.4 and

0.7 um): “Laser Radiation – Do Not Stare into Beam or View Directly with Optical Instruments”.

8.2.3 All other Class 3R lasers or laser systems: “Laser Radiation – Avoid Direct Eye Exposure”.

8.2.4 Class 3B lasers or laser systems: “Laser Radiation – Avoid Direct Exposure to Beam”.

8.2.5 Class 4 lasers or laser systems: “Laser Radiation – Avoid Eye or Skin Exposure to Direct or Scattered Radiation”.

8.3 Labeling of Protective Equipment.

8.3.1 All eyewear must be clearly labeled with the optical density and wavelength. Color-coding or other distinctive identification is recommended in multi-laser environments.

8.3.2 All laser protective windows must be labeled with the optical density and wavelength(s) for which protection is afforded, and should be labeled with the threshold limit and exposure time for which the limit applies, and the conditions under which protection is afforded.

8.3.3 All laser protective barriers must be labeled with the barrier threshold limit and exposure time for which the limit applies, and beam exposure conditions under which protection is afforded. Contact the LSO for assistance.

9.0 Training.

9.1 Laser Safety Training. All faculty, staff, and students who wish to operate Class 3B and Class 4 laser systems at GHSU shall:

9.1.1 Read this Laser Safety Guide, and

9.1.2 Complete the on-line laser training module.

9.2 Laser System-Specific Training.

9.2.1 The PLU is responsible for conducting and documenting laboratory-specific safety training for laser users, including a thorough review of the laser equipment, administrative and engineering controls, and alignment and standing operating procedures.

9.2.2 Retraining is required whenever a new laser hazard is introduced into the work area.

9.2.3 Users of class 3b and 4 lasers will complete refresher on-line training annually at our Radiation Safety web site as shown below:
(<http://georgiahealth.edu/services/ehs/radsafe>)

10.0 Medical Surveillance. Baseline laser ophthalmologic exams are required for Class 3B and Class 4 laser users when designated by the LSO. Contact Occupational Health Services at 706-721-3418 to schedule an appointment.

11.0 Special Considerations.

11.1 Service of Embedded Lasers. Access to Class 3B or Class 4 lasers or laser systems enclosed within a protective housing or protected area enclosure is limited to properly trained service professionals and by specific engineering and administrative controls. Contact the LSO for more information.

11.2 Outdoor Controls. Projection of Class 3R, 3B, or 4 laser beams in the outside environment requires prior review by the LSO and concurrence by the LSS.

11.3 Laser Pointers are typically Class 2 or Class 3R devices. They are usually hand held and emit a low-divergence visible beam of less than 5 milliwatts. Laser pointers are usually limited to a Class 3R (5 mW) output or less; however, there are more powerful laser pointers available and these devices present a significant potential for an eye injury if viewed directly. Laser pointers are typically used in lecture presentation to identify objects or images. Safety recommendations include:

11.3.1 Never point a laser pointer at anyone, and never look directly into the beam.

11.3.2 Never aim a laser pointer at surfaces that would reflect the light back, such as mirrors or mirrored surfaces.

11.3.3 Purchase only those laser pointer devices with a clear warning on the label about the potential to cause eye damage. Read the instructions carefully, and follow them.

11.3.4 Choose a laser pointer that stays “ON” only when you apply pressure with your fingers. That way you can never leave the beam “ON” by accident.

11.3.5 Choose laser pointers with power that is appropriate to the intended purpose.

12.0 Laser Accidents.

12.1 Immediate response. For health-threatening injury, including eye or skin injuries, call 1-2911 or proceed directly to the hospital emergency room..

12.2 Accident Investigation.

12.2.1 The LSO investigates laser accidents and reports findings to the LSS. The investigation includes:

12.2.1.1 Interviews of the injured worker, the PLU, and any witnesses.

12.2.1.2 An examination of the work place to identify contributing factors.

12.2.1.3 A determination of the cause of the accident.

12.2.1.4 Recommendations for corrective actions.

12.2.2 The LSS will review the accident investigation report and determine when and under what conditions laser operations may resume.

12.2.3 The PLU will not resume laser operations until authorized by the LSO.

13.0 Record Keeping. The Radiation Safety Office shall retain records of Laser Safety Subcommittee meetings, laser system registrations, laser system inventories, laser safety guides, and site inspection reports, for three years before retiring the records in accordance with GHSU guidelines.